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Design and analysis of lower control arm of suspension system

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ABSTRACT

In automotive suspension, a control arm is a hinged suspension link between the chassis and the suspension or hub that carries the wheel. The chassis end of a control arm usually rubber busing, is attached by a single pivot. It controls the position of the tyre end in only a single degree of freedom, maintaining the radial distance from the chassis mount.

The single bushing does not control the arm from moving back and forth; this motion is constrained by a separate link or radius rod. Wishbones are triangular and have two widely spaced chassis bearings, which constrain the tyre end of the wishbone from moving back and forth, controlling two degrees of freedom, and without requiring additional links. Most control arms form the lower link of a suspension with few designs using them as the upper link, usually with a lower wishbone. The lower arm should be sturdy, in many cases reported by the users of a particular vehicle, fluttering noise heard over the humps.

This paper calculates the forces acting on the lower arm of a four-wheeler with critical loading conditions as the Finite Element Analysis carried for the McPherson type suspension system. The static analysis performed to determine the location of maximum deflection and stress distribution while the vehicle is stationary and moving over the hump at different speeds. The free-free and constrained modal analysis performed on the suspension arm to find the natural frequency. The fatigue analysis performed to find the life of the arm. Further, by performing topological optimization leads to reduction in the weight.

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1. Introduction

Vehicles ranging from bullock carts to modern vehicles use suspension system such as, leaf springs, pneumatic and hydraulic systems. The main aim of using a suspension system is to prevent the shocks reaching the body of the vehicle from the rough roads and ensure that the vehicle's wheels remain in contact with the road. There are two types of forces acting on the suspension system of a vehicle, the first is the force from the road disturbance, and the second is the load disturbance. Road unevenness could be of high magnitude and low frequency (such as mountains) and the small magnitude and large frequency (because of rough roads). Load disturbance includes the forces induced by the changing acceleration, braking and cornering. A suspension system should respond very smoothly against road disturbances and should be robust against load disturbances.

The suspension system consists of spring, damper and strut. The function of spring is to absorb the shocks and store the energy, but

damper (shock absorber) is required to dissipate it. Many types of springs are used in the automobile suspension system like leaf spring, helical coil spring, torsion spring, air spring, etc. Mostly hydraulic or pneumatic dampers are used in the car suspension having a piston and a rod with fluid inside it, which slows down the spring motion.

After conducting analysis like dynamic, fatigue and kinematic analysis many modification carried out on the suspension system by Saurabh D. et al. [1]. However, it's observed that no study was conducted on the failure of strut. J. A. Tamboli et. al. [2] reported that the Power Spectrum Density (PSD) of a particular road decreases exponentially with respect to time.

In addition, they found the effect of changing the speed and wheelbase based on the Root Mean Square Acceleration Response (RSMAR) values. Loprencipe, G., Zoccali, P. [3], discussed about the road roughness by using validated artificial road profiles generated using the hanning window. Lu Sun [4] stated that power spectral density (PSD) values used for obtaining the road profile. Further, found the international roughness index (IRI) considering the PSD of road, which used for evaluating pavement surface roughness.

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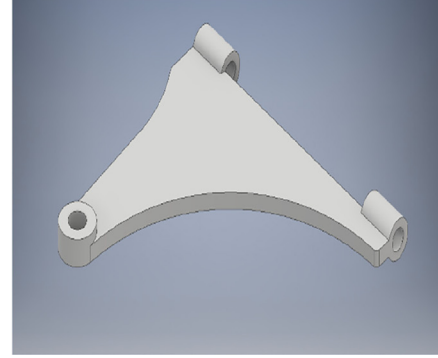
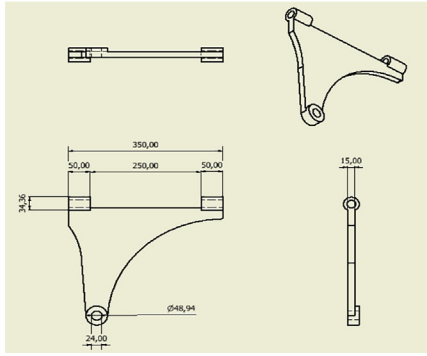


Fig. 1. Dimensions of lower control arm and final CAD model.

Mothiram K. Patil and M. S. Palanichamy [5] used the concept of lumped mass for the mathematical model of the human being. This model is then analysed for the vertical vibration response for the new seat, which made for reducing the transmission of shocks. Steven Roy and Zhaocheng Liu [6] made a two dimensional Mathematical model of vehicle having restricted motion in eight DOF. They performed simulation for different cases and the effect of wind flow was considered by CFD analysis. G. Verros, C. Padimitriou and S. Natsiava [7] work focussed on optimisation of quarter car models under road excitations. Lu Sun, Xi Cai [8] demonstrated method to optimise the forces acting on the vehicle by using genetic algorithm and proved the value of sprung mass, unsprung mass, spring stiffness and damping coefficient will reduce the vibrations in the automobile. Masih Sharifi et al. [9] proposed the multiobjective genetic algorithm to optimise the vibrations in the two DOF automobile model.

Xiaobin Ning et al. [10] presented the study on the vibration analysis of suspension system using the ADAMS tool. The study aimed to improve the comfort of passengers isolating them from any external shocks. Özgür Bayrakdar [11] worked on random vibration in automobile. The study is on the vibration analysis of different suspensions and the response equations to the random vibration on different models having different DOFs. Shahrul Abdullah et al. [12] conducted a numerical analysis on the lower control arm of automobile and calculated the damage caused by the vibrational fatigue. Performed the stress analysis and Eigen value analysis by using numerical techniques. Brian Baillargeon, Senthil Vel [13] presented a paper in which they analysed the suspension system on ABAQUS to find the behaviour of suspension against the random road profile considering the suspension as a cantilever beam.

According to the available literature the main problems arising due to loading conditions and vibrations stated as follows;

- Occurrence of resonance, avoided by changing the natural frequency through either increase of the mass of components or through changing the stiffness. Both the cases tend to increase the stresses in lower arm.
- Obtaining the loading conditions, force imposed by the road to the vehicle considered, which depends on many factors like the damping coefficient of suspension, stiffness of the suspension, and tire stiffness. Some other key factors like varying speed and acceleration of car can be considered
- Fluctuating forces acting on lower arm due to vehicle undergoing random vibrations and calculating the life of suspension system

In the work presented analyses of the lower arm to determine the stresses induced and deformation using the FEM performed.

Modal analyses carried out to find the resonance conditions. Finally, fatigue analysis carried out to make the life estimation of lower arm.

1.1. Part design of lower arm

The dimensions of the front McPherson suspension system are physically measured and the part design carried out using Auto-desk Inventor (Refer Fig. 1). The lower arm connected to the car chassis on one side while on the other side; its connected to the tyre. One end has to withstand the force imposed by the road whereas on the other side it has to take the weight of the car. The thickness of the plate used in lower arm modelling is of 15mm thick.

Fig. 1. Dimensions of lower control arm and final CAD model

1.2. Meshing

Before meshing, using the de-feature panel in Hypermesh initially the model is checked for the features (unnecessary features are deleted or which complicate the meshing). The 3D meshing carried out using the brick HEXA elements and PENTA elements as shown in Table 1. The meshed model is show in Fig. 2.

Hexa 8: 42,586 elements and Penta 6: 366 elements (Table 1).

1.3. Material and its properties

The lower arm of suspension system is made of aluminium. A357 Alloy mainly used for the production of lower arm and many other automobile parts because when it is heat-treated it becomes one of the strongest sand cast aluminium alloys mentioned in Table 2.

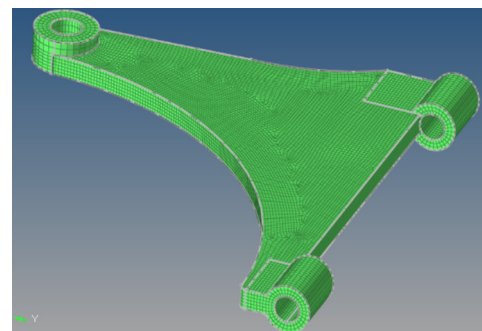


Fig. 2. The meshing of the lower control arm.

Table 1

Quality Parameters considered for the meshing.

1	Minimum element length	< 1.875mm
2	Average element length	= 2.5mm
3	Maximum element length	> 5mm
4	Warpage	> 5mm
5	Aspect ratio	> 5
6	Skewness	> 60
7	Jacobian	< 0.7
8	Maximum quad angle	> 135
9	Minimum quad angle	< 45
10	Maximum tria angle	> 120

Table 2

Material properties.

Properties	Aluminum A357
Elastic modulus	70GPa
Elongation at break	3.70%
Fatigue strength	100MPa
Poisons ratio	0.33
Shear modulus	26GPa
Shear strength	240MPa
Tensile strength (ultimate)	350MPa
Tensile strength (yield)	270MPa
Density	2.6g/cm ³
Brinell hardness	100

2. Calculations of load acting on lower control arm

The forces acting on the lower arm based on the regular road conditions are

- Braking force mainly comes into existence when brakes are actuated.
- The frictional force that acts on every member as friction is evident in every machine component.
- A vertical force that decides the maximum vertical force that it can carry.
- The cornering force acting in a tangential direction when vehicle takes turn
- Hump force that acts on the lower arm when the wheels undergo rapid motion over it.

2.1. Load acting on the lower arm when the vehicle is stationary

First, we have to find the longitudinal centre of gravity of the car.

Distance from the front axle to the centre of gravity = 1222.38mm

Distance from the rear axle to the centre of gravity = 2520mm – 1222.4mm = 1297.6mm

Weight of the car acting at the centre of gravity = mass of the vehicle × gravity

$$= 1340 \times 9.81 = 13145.4N$$

Weight of the car acting on the front axle = 6768.83N

Weight of the car acting on each wheel = 3384.4N

A load of 3384.4N acts on the lower arm of the car when the car is stationary.

2.2. Load acting on the lower arm when the vehicle is moving over the hump

When the car moves over the hump, we have to take the braking force, cornering force, and centrifugal force from the wheels.

Braking force occurs due to the actuation of brakes. Cornering force acts vertically when the car takes a turn, and a centrifugal force acts in a horizontal direction due to the wheel's fast rotation over the road. Consider the vehicle is moving at a speed of 15km/hr that is equal to 4.16m/sec passing over the hump.

2.3. Braking force

Longitudinal weight transfer = 62.1kg

Vertical load due to braking on the wheel (A) = 1624.5N

Frictional force = 1624.5 × 0.8 = 1300N

Force upright is half of the frictional force = 650N

Moment = 650 × 0.076 = 49.38Nm

2.4. Cornering force

Lateral load transfer

$$= \frac{\text{acceleration} \times \text{c.g. height} \times \text{m} \times \text{wt distribution}}{\text{g} \times \text{track width}} = \frac{5.84 \times 0.5 \times 345 \times 0.63}{9.81 \times 1.2192} = 53.06\text{kg}$$

Vertical load due to cornering on the wheel (B) = 1586.62N

Frictional Force = 1586.62 × 0.8 = 1269.3N

Moment due to friction = 1269.3 × 0.09 = 114.23Nm

Force at upper arm

$$= \frac{114.23}{0.0762} = 1500N$$

2.5. Total force when the vehicle is moving over the hump

Hump force of arm

$$= \frac{(\text{centrifugal force} + A + B) \times C}{D}$$

A: Vertical force while braking,

B: Vertical force while cornering,

C: Distance from roll case to wheel centre

D: lower length (in side view)

Centrifugal force

$$= \left\{ \frac{(\text{mass} \times \text{weight distribution}) \times \text{velocity}^2}{\text{bumpradius}} \right\} = \left\{ \frac{(345 \times 0.63)}{1} \times 4.16^2 \right\} = 1880.68N$$

Hump force

$$= \left\{ \frac{(1880.68 + 1624.5 + 1586.62) \times 0.4064}{0.281} \right\} = 7364.08N$$

2.6. Considering that only half of the load, gets transmitted

Hump force = 3682.04N

When the vehicle moves at a particular speed over the hump, the force acting on the lower arm is also calculated. Total hump force acting on the lower arm shown in the table

3. Simulation and results

3.1. Displacement and stress plot concerning the vehicle weight alone

In Fig. 3, the total displacement that the lower arm undergoes as shown. 3384N The load applied at one end of the lower arm, and other ends kept fixed. The maximum displacement found to be 3.425mm after performing the simulation.

Fig. 4 shows the maximum Von Mises stress induced in the lower control arm when the load 3384N is applied. The maximum stress that can be seen from the result is 176.3MPa.

3.2. Stress plot concerning the vehicle moving over the hump

Fig. 5 shows the maximum Von Mises stress induced in the lower control arm when the vehicle moves over the hump. The

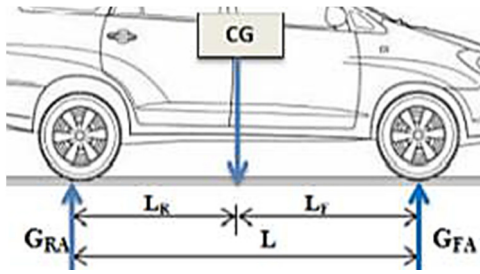


Fig. 3. Longitudinal gravity of the car.

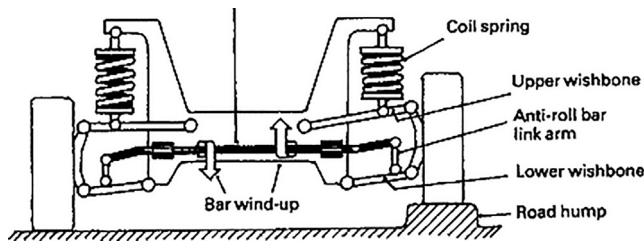


Fig. 4. Wheel of the car over the hump.

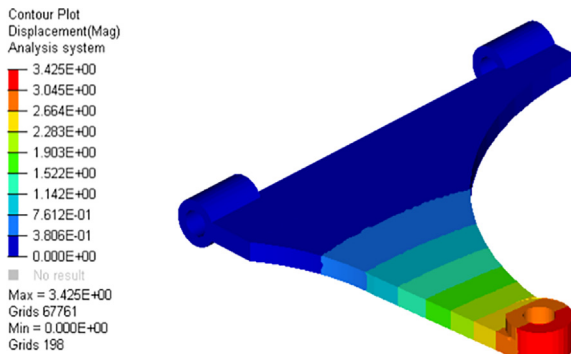


Fig. 5. DISPLACEMENT when the vehicle is stationary.

maximum stress that can be seen from the result is 208.4MPa (Fig. 6).

3.3. Effect of change in velocity of the vehicle on the stresses in the lower arm

The effect of change in vehicle velocity on the variations in stress, we have to plot the graph. Velocity plotted on the x-axis, whereas the stresses plotted on the y-axis.

From the above plot, we can see that higher stresses induced in the lower arm when the vehicle travels over a hump at a higher velocity. Hence, there are higher possibilities of the arm's failure when the vehicle passes over the hump at a very high speed. When the vehicle is moving over the hump at the speed of 30km/hr , Von Misses stress in the arm is 250MPa . When we compare this stress value with the ultimate strength of aluminum A357, we get safety factor of the value 1.4. When the vehicle goes over the hump at speed, more than the 70km/hr stresses generated in the arm are above 350MPa (ultimate strength of the material), and hence the arm will break in such unlikely situations.

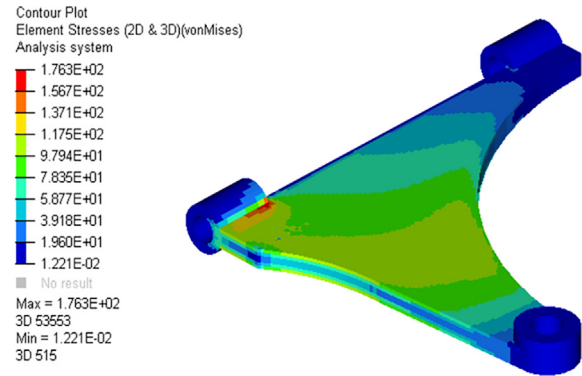


Fig. 6. Von misses stress plot when the vehicle is stationary.

Table 3

Forces arising at different velocities.

Velocity (km/hr)	Hump force (N)
10	3156.3
15	3682.4
20	4376.2
30	4812.0
50	5726.0
60	6573.3
70	7231.6
75	7682.3

3.4. Mode shapes for free-free analysis

In free-free modal analysis, we get 10 modes of vibration within the frequency range 0Hz to 2000Hz . Different modes summarized in the Table 3.

Table 3 shows that the first six natural frequencies of the control arm under free-free modal analysis are zero. These first six natural frequencies represent the motion of the arm in six independent degrees of freedom. The next four natural frequencies range in between 528Hz to 1536Hz .

3.5. Constrained modal analysis

In this case, the two ends of lower arm are fixed, and modal analysis is performed. For instance the first mode shape shown in fig below having the range of 0Hz to 3495Hz and similarly other mode shapes have been tabulated in Table 4 and Table 5

From Table 4, we can observe that the arm's first 10 natural frequencies, when fixed at the two ends, range between 349Hz and 3090Hz . The natural frequency of the car suspension ranges between 1Hz to 2Hz . Since the lower arm does not have

Table 4

Natural frequency for free-free analysis.

Mode	Frequency	Eigen Value	Stiffness
1	$6.1357e^{-3}$	$1.4862e^{-3}$	$1.4862e^{-3}$
2	$6.3790e^{-3}$	$1.6064e^{-3}$	$1.6064e^{-3}$
3	$6.4445e^{-3}$	$1.6396e^{-3}$	$1.6396e^{-3}$
4	$6.6775e^{-3}$	$1.7601e^{-3}$	$1.7601e^{-3}$
5	$6.7759e^{-3}$	$1.8125e^{-3}$	$1.8125e^{-3}$
6	$6.8265e^{-3}$	$1.8397e^{-3}$	$1.8397e^{-3}$
7	$5.2865e^2$	$1.1031e^7$	$1.1031e^7$
8	$8.0863e^2$	$2.5814e^7$	$2.5814e^7$
9	$1.4088e^3$	$7.8362e^7$	$7.8362e^7$
10	$1.5366e^3$	$9.3224e^7$	$9.3224e^7$

Table 5
Natural frequency for constrained modal analysis.

Mode	Frequency	Eigen value	Stiffness
1	3.4952e ²	4.8229e ⁶	4.8229e ⁶
2	6.3886e ²	1.6113e ⁶	1.6112e ⁷
3	7.1020e ²	1.9912e ⁷	1.9912e ⁷
4	8.3453e ²	2.7494e ⁷	2.7494e ⁷
5	1.2880e ³	6.5492e ⁷	6.5493e ⁷
6	1.6460e ³	1.0645e ⁸	1.0696e ⁸
7	2.2195e ³	1.9465e ⁸	1.9448e ⁸
8	2.3149e ³	2.1156e ⁸	2.1156e ⁸
9	2.4668e ³	2.4536e ⁸	2.4023e ⁸
10	3.0906e ³	3.7712e ⁸	3.7710e ⁸

Table 6
Results for original and optimized model.

Content	Original Model	Optimized Model
Maximum Von-Mises Stress at 30km/hr	250MPa	221MPa
Failure occurs at	74km/hr	70km/hr

Table 7
Mass comparison of initial and optimised model.

Model	Mass (Kg)	Cost (Rs.)
First model	2.017	294.46
Optimised model	1.611	233.59
% mass reduction	20.182	59

Table 8
Deflection value for theoretical and analytical method.

Deflection in the arm	Theoretical	FEA	% error
	3.49 mm	3.425 mm	0.018

any natural frequency in the range, **1Hzto2Hz** there is no resonance possibility.

3.6. Fatigue analysis

Fatigue life plot and damage plot of the lower arm is obtained by performing fatigue analysis. The stress life approach is used for calculating life. The load history was given as per the Altair motion solve simulation results, and Gerber criteria were used for calculating life. Total fatigue life and damage are obtained using the stress life approach, as shown in Figs. 7 and 8.

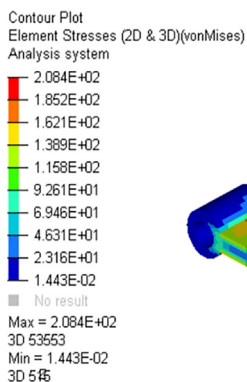


Fig. 7. Stresses in the arm when the vehicle passes over the hump at 15km/hr

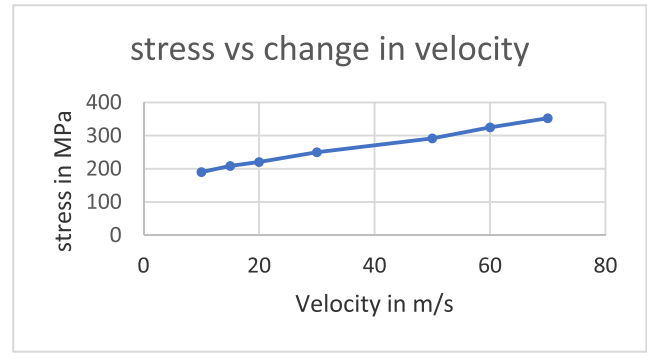


Fig. 8. The plot of the velocity of a vehicle vs. the stresses in the lower arm.

The maximum damage of 0.0014 is recorded in the simulation. Besides, the life of the model found to be $1e^{20}$ cycles. According to the Miner Palm grain cumulative damage, theory fracture occurs when the damage is more significant than one for the fatigue cycle. Maximum damage was found as 0.0014, which is less than one; hence the arm can withstand the stresses without getting a crack Shown in Table 6.

4. Topology optimization

Topology optimization is to control the objective functions such as weight, volume, etc., while ensuring that it meets the constraints, i.e., strength, natural frequency, etc. In this paper, the focus will be on reducing the weight while increasing the strength of the lower arm (thickness of the arm.). By giving appropriate constraints (stress values), the optimization carried out using analysis software (Optistruct) Shown in Table 7.

4.1. Geometric model of the optimized control arm

Thickness in the critical region increased from 15 mm to 20 mm, which increased the strength of the arm and the material from the low stress region removed, as the weight reduced to 1.61 kg as compared to earlier model, which was 2.07 kg. The CAD model of the optimized arm depicted in Fig. 9.

4.2. Displacement and stress plots of the optimized model

In Fig. 10, the total displacement that the lower arm undergoes shown when the vehicle is stationary. 3384N load is applied at one end of the lower arm, and other ends are kept fixed. The maximum displacement found to be 1.57mm after performing the simulation. Fig. 11 shows the maximum Von Misses stress induced in the control arm when the load of 3384N is applied. The maximum stress that is seen from the result is 159.6MPa. (Fig. 13.Fig. 14.Fig. 15. Fig. 16) Shown in Table 8.

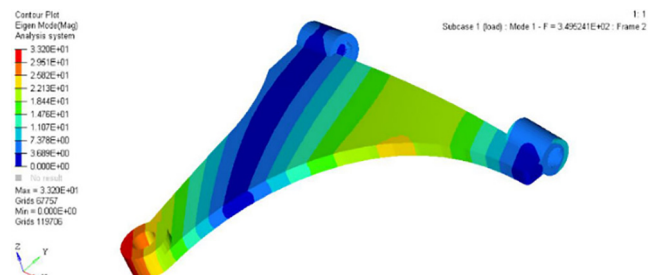


Fig. 9. Mode 1.

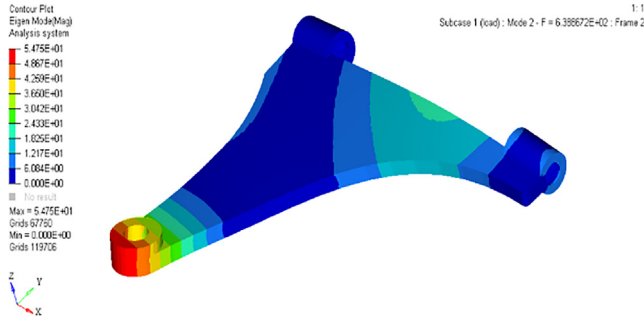


Fig. 10. Mode 2.

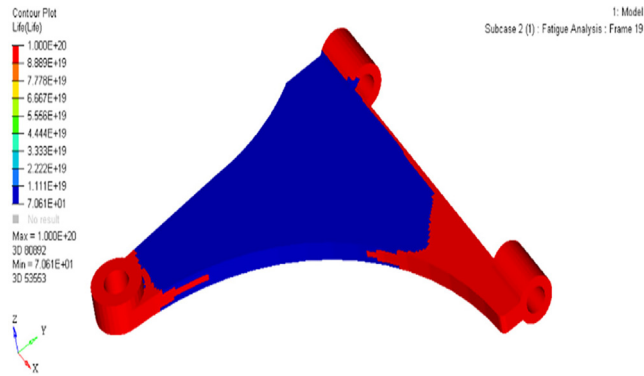


Fig. 11. Life plot of the lower arm

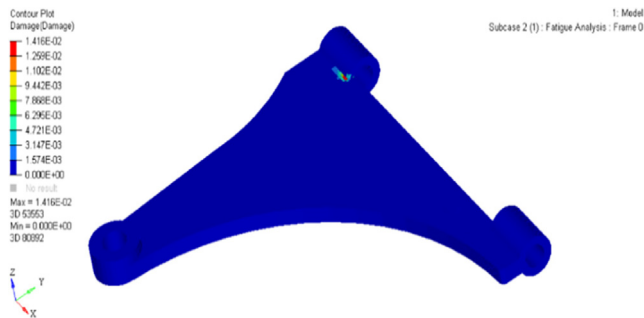


Fig. 12. Damage plot of the lower arm

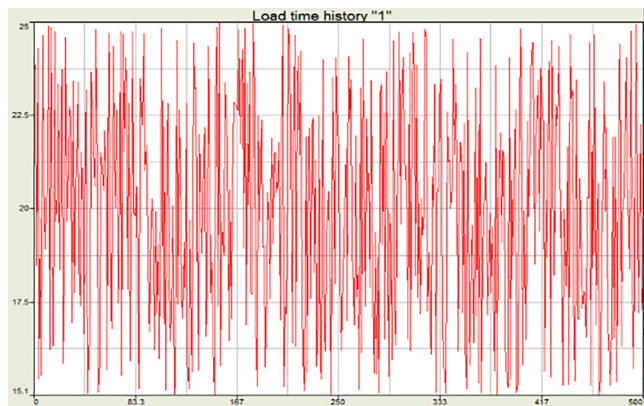


Fig. 13. Load time history plot for the lower arm

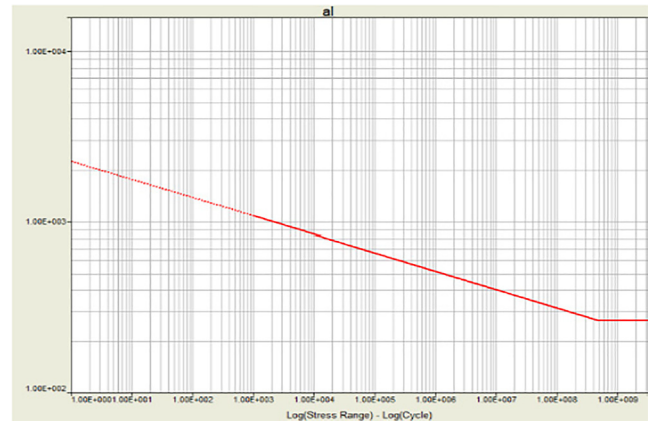


Fig. 14. SN curve for aluminium A357.

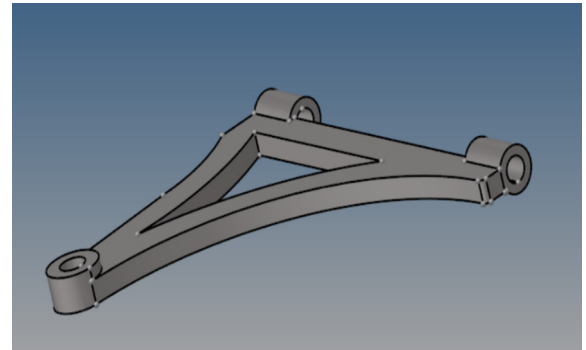


Fig. 15. CAD model of the optimized model.

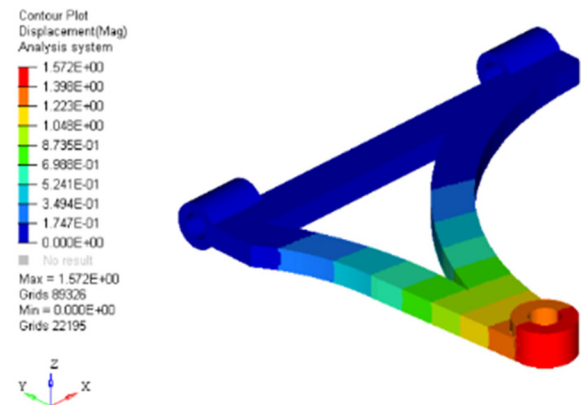


Fig. 16. Displacement plot of lower arm when the vehicle is stationary.

4.3. Comparison of stresses of the optimized model with the old model at different velocities

By plotting the stresses obtained from the graphs, we can deduce that significant reduction of weight in the lower arm and further reduction of stresses

Fig. 18 stress plot for comparison between the old model and the optimized model (Fig. 20.Fig. 21.Fig. 22).

By doing optimization, as shown in the table below, we were successful in reducing weight as well as increasing the factor of safety.

By performing topology optimisation, we were able to reduce the weight of the first model by 0.4Kg. The cost of aluminium in India is around 145Rs/Kg.

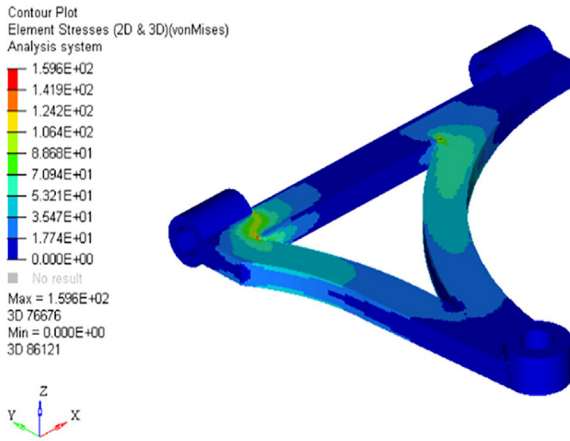


Fig. 17. Stress plot of the optimized arm when the vehicle is stationary.

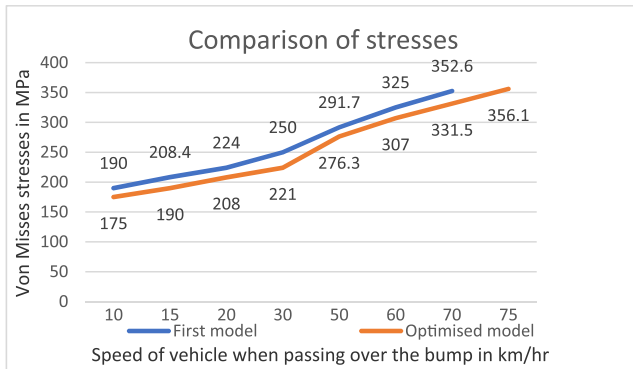


Fig. 18. Stress graph for comparison between the old model and the optimized model.

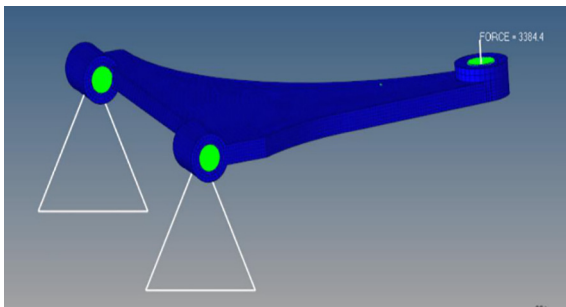


Fig. 19. Stresses in the arm when the vehicle passes over the hump at 15 km/hr.

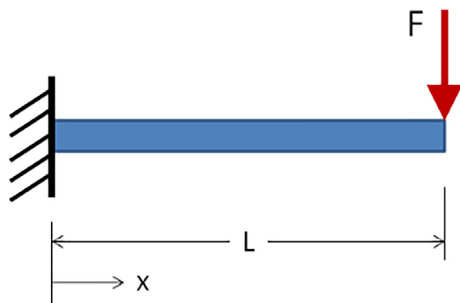


Fig. 20. The figure of the velocity of a vehicle vs. the stresses in the lower arm via cantilever beam.

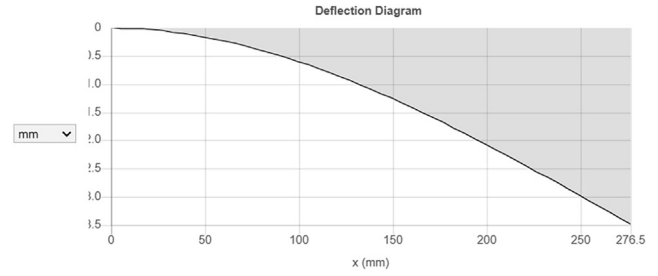


Fig. 21. Deflection of a cantilever beam.

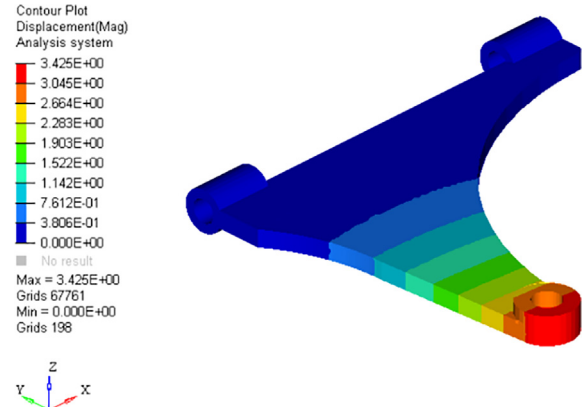


Fig. 22. Displacement plot by simulation.

5. Validation of the results

As the control arm fixed on one end while loaded on the other end, assumed as cantilever beam with a tapered cross-section.

Moment of inertia

$$I = \frac{b \times d^3}{12} = \frac{100 \times 10^{-3} \times (15 \times 10^{-3})^3}{12} = 8.43 \times 10^{-8}$$

Deflection

$$= \frac{F \times L^3}{3 \times E \times I} = \frac{3384 \times (276.5 \times 10^{-3})^3}{3 \times 8.43 \times 10^{-8} \times 81000} = 3.492 \text{ mm}$$

The maximum displacement from the FEA analysis is 3.425 mm at the end where the load is applied. The theoretical calculation gives identical displacements; hence, the simulations performed on the hyper-mesh opistrukt tool are accurate enough.

5.1. Fatigue analysis

When the vehicle is moving over the uneven road surface dynamic loads act on the arm. Under Such type of repeated dynamic load, the component tends to fail below its ultimate tensile strength. The limit at which it fails called as endurance limit. Fatigue analysis performed in order to find the life of the control arm when the arm subjected to constantly varying loads.

5.2. Endurance limit

Aluminium is malleable and can deform to take different shape it can withstand more cycles than most of the ferrous brittle material. Endurance limit of aluminium A357 calculated by taking into account all the correction factors.

$$S'_e = 0.4 S_{ut} = 0.4 \times 350 = 140 \text{ MPa}$$

$$\text{Surfacefactor} = K_a = 0.968$$

$$\text{Sizefactor} = K_b = 0.886$$

$$\text{Loadfactor} = K_c = 1$$

$$\text{Temperaturefactor} = K_d = 1$$

$$\text{Stressreductionfactor} = K_e = 0.852$$

$$S_e = S'_e K_a K_b K_c K_d K_e = 140 \times 0.968 \times 0.886 \times 1 \times 1 \times 0.852 = 102.3 \text{ MPa}$$

Endurance limit of aluminium is **102.3MPa** considering all the correction factors. The maximum stress induced in the arm because of the variation in road is **25MPa**. This value is much smaller than the endurance limit hence the material can withstand infinite cycles for this load time history.

5.3. Simulation test

When the problem is solved using finite element analysis on analysis software

Maximum displacement from the FEA analysis is **3.425mm** at the end where the load is applied. In addition, the theoretical calculation gives the identical displacements hence the simulations performed on the analysis software are accurate enough.

6. Conclusion

As the car passes over the hump, the braking force, cornering force, centrifugal force, and reaction forces are analysed when the car was static and while moving over the humps. Stresses acting on the arm found at different speeds at which vehicle passes over the hump. During the static analysis, when the vehicle was stationary, the displacement was found to be **3.425mm**, and von misses stress acting on the lower control arm was **176.3MPa**. In free-free modal analysis and constrained analysis were carried and results were satisfactory and there is no possibility of resonance. The fatigue life analysis using SN approach, the life estimation been found out. By performing topology optimization, considerable saving of material is been observed. From this observation, we can conclude that the optimized model was lighter and safer than the initial model. In addition, it was clear from fatigue analysis that the arm could withstand the loads for 10^{20} cycles without getting a crack.

CRedit authorship contribution statement

M. Sadiq A. Pachapuri: Visualization, Writing - original draft, Writing - review & editing. **Ravi G. Lingannavar:** Methodology, Resources, Supervision. **Nagaraj K. Kelageri:** Formal analysis, Validation. **Kritesh K. Phadate:** Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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